

1. What does image classification mean? Explain with one example.

Image classification is the process of assigning an image to a predefined category or label using machine learning or deep learning models.

Example:

A model is trained to classify images of clothing items into categories such as:

- T-shirt
- Trouser
- Sneaker
- Bag

If an image of a sneaker is given as input, the model predicts the label sneaker.

2. Why is using a Dense Network not always a suitable choice for images?

A dense network treats every pixel as an independent input and ignores the relationship between another pixels.

There are some problems with this. they are

- Requires a very large number of parameters.
- High Computational cost.
- More prone to overfitting.

CNNs are preferred because they can automatically learn visual features while using fewer parameters.

3. In a CNN, what is convolution, and what does a filter or kernel do?

Convolution is a mathematical operation where a small matrix (filter/kernel) slides over the image and performs element-wise multiplication and summation.

A Filter detects specific patterns like edges, corners, textures

For example, an edge-detecting filter highlights boundaries in an image.

4. What is a **feature map** in a convolutional neural network?

A feature map is the output produced after applying a filter to an image.

It shows where in the image a particular feature like edge, texture, patterns were detected.

Example: If a vertical-edge filter is applied, the feature map will highlight all vertical edges in the image.

5. Why is **weight sharing** used in CNNs? Explain.

It means that the same filter with same weight is applied across the entire images.

It is used to Reduces number of parameters, detects features anywhere in the image and Prevents overfitting.

6. How does the **ReLU activation function** work? Write the ReLU output for the following values:

[-3, 0, 2, 5]

ReLU stands for **Rectified Linear Unit**, and it transforms an input value x using:

$$\text{ReLU}(x) = \max(0, x)$$

In the given values

[-3,0,2,5]

Apply ReLU element-wise:

- $\text{ReLU}(-3) \rightarrow 0$ negative value becomes 0
- $\text{ReLU}(0) \rightarrow 0$ value it stays 0
- $\text{ReLU}(2) \rightarrow 2$ positive value stays same
- $\text{ReLU}(5) \rightarrow 5$ positive value stays same

Final Output:

[0, 0, 2, 5]

7. What is max pooling, and why is it used in CNNs?

Max pooling takes a small window and outputs the maximum value from that region.

It is used in CNN because Reduces image size, keeps important features and Provides translation invariance.

Example: 2×2 region: [1, 5; 3, 2]

The max pooling output is 5.

8. In the Fashion-MNIST code, why are images reshaped from 28×28 to 28×28×1?

CNNs expect images in the format:

Height × Width × Channels

Fashion-MNIST images are grayscale. It is **1 channel**.

So, 28×28 becomes:

28 × 28 × 1

This tells the CNN that each pixel has one intensity value .

9. What is the difference between training data, validation data, and test data?

Training data is the portion of the dataset used to teach the model. This is where the model learns patterns, adjusts its weights, and improves its predictions

Validation data is separate from training data and is used during the training process to tune hyperparameters, evaluate how well the model generalizes, and detect issues like overfitting.

This model does not learn from validation data, it only uses it for performance checks while training.

Test data is used after the model has been fully trained and finalized. It provides an unbiased evaluation of how the model performs on completely unseen data, giving a realistic estimate of how well it will work in real-world situations.

10. What is **overfitting**? If training accuracy is high but validation accuracy is low, what does this indicate?

Overfitting happens when a model learns the training data too deeply, including patterns that are not actually useful such as noise, random fluctuations, or very specific details that don't generalize to new situations.

As a result, the model performs extremely well on the data it has already seen but struggles when faced with new, unseen examples.

So, when we observe **high training accuracy but low validation accuracy**, it means the model has essentially memorized the training data instead of learning the underlying patterns. This gap between training and validation performance is a clear sign of overfitting, showing that the model cannot generalize well and will likely perform poorly in real-world scenarios.